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Exchange rate intervention in small open economies: The role of risk premium and commodity price shocks

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ABSTRACT

We study the effect of country risk premium and commodity price shocks on monetary policy in small open economies. Risk premium shocks not only can explain most of the variance in the exchange rate, but also can expand GDP. Our estimations indicate that the impact of a real depreciation on exports far exceeds not only the balance sheet effect, but also the effect of an increase in the cost of imported inputs. However, in a more complex environment, where the changes in exports are offset by negative shocks to commodity prices, the contraction stemming from the balance sheet effect reappears.

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"Central banks in small open economies should openly recognize that exchange rate stability is part of their objective function."International Monetary Fund[1]

1. Introduction

The design of monetary policy in small open economies poses important challenges that are not present in developed countries. Small open economies must continuously deal with volatility in international financial markets and international trade, especially from the high variability of country risk premiums and commodity prices, which could push the central bank to change its monetary stance. In this study, we also include other shocks that have been proposed as determinants of the exchange rate (such as productive and monetary shocks).[2]

The real exchange rate is one of the key variables through which fluctuations in international markets are transmitted to domestic economies. For example, unexpected external shocks that alter the exchange rate may increase the cost of the external debt service, the value of income from commodity exports, the cost of imported inputs, and so on. Thus, the change in the real exchange rate may alter the expected path of inflation, forcing central banks to adjust their monetary policy.

Much of the literature on monetary policy in open economies has focused on whether central banks respond to the real exchange rate. The evidence from empirical studies indicates that many countries include the real exchange rate in their policy reaction function. The evidence is not conclusive, however: countries like Australia and New Zealand do not incorporate the exchange rate in their policy reaction function (Lubik & Schorfheide, 2007).

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[1] Quoted in Blanchard, Dell'Ariccia, and Mauro (2010).
[2] In Appendix A, we define all the shocks considered in this study.

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On the other hand, welfare analysis has produced contradictory results depending on the model proposed (Bergin, Shin, & Tchakarov, 2007). For example, Ball (1999), Svensson (2000), and Batini, Harrison, and Millard (2003) find that including the real exchange rate marginally improves the macroeconomic performance of central banks. In contrast, studies such as Wollmershauser (2006), Morón and Winkelried (2005), and Cavoli (2009) show that defending the exchange rate may be useful in a context of financial instability or as a response to fear of floating.

Our first goal is to estimate a simple model that is sufficiently general to incorporate a rich variety of shocks for explaining the real exchange. We build a dynamic stochastic general equilibrium (DSGE) model for a group of small open economies (namely, Australia, New Zealand, Chile, Colombia, and Peru). We consider this group of countries because they are all small open economies, inflation targeters, and commodity exporters, and they have frequently been hit by shocks that change the conditions for accessing international financial markets and the prices of their main exports (commodities).

The model considers imperfect capital markets (in which the country risk premium depends on the ratio of external debt over GDP), restricted consumers, the balance sheet effect of exchange rate devaluations, imported inputs, commodity exports, imperfect pass-through of the exchange rate, and wage indexation. Finally, we use Bayesian techniques so that we can estimate all the equations and shocks simultaneously.

Our second target is to establish whether further exchange rate intervention is necessary in these small open economies. To do so, we use the allocations of the Ramsey problem as a benchmark. We obtain this by minimizing an arbitrary loss function in term of the variance of the inflation rate, output gap, and monetary policy rate. This assumption is for the sake of simplicity, since we do not want to characterize optimal monetary policy given a specified utility function. Instead, we explore the more empirical question of how central banks can improve their reaction to a risk premium shock in the face of any objective function that incorporates inflation and output gap fluctuations.

The results of the paper are the following. First, the risk premium shocks cannot only explain most of the variances of the exchange rate but can also cause important reallocation of resources across sectors in the short run: a positive risk premium shock expands GDP. Other shocks, such as the commodity price shock, play only a minor role. We define risk premium shocks as unanticipated changes in credit risk conditions related to external debt. This type of shock directly reduces the resources that families have to smooth consumption across time because they must pay a higher interest rate. In other words, in our study a positive risk premium produces a credit spread between the real cost of foreign borrowing and the foreign interest rate. Nevertheless, the real depreciation increases exports and therefore GDP.

This last result is controversial. The Mundell–Fleming–Dornbusch model indicates that a greater devaluation of the domestic currency is necessary to adjust the economy in response to a negative external shock. However, this is at odds with the balance sheet effect or the so-called original sin. Our model estimation indicates that the impact of a depreciation of the domestic currency on exports far exceeds not only the balance sheet effect, but also the effect of an increase in the cost of imported inputs that could cause a *J* curve in the short run.

Second, from the perspective of welfare analysis, our results do not support the prescription of the Mundell–Fleming–Dornbusch model. In the case of a risk premium shock, the impulse response function shows that both the inflation rate and the growth rate increase simultaneously as a result of a real depreciation. Central banks can thus avoid this excess volatility by raising the interest rate. We show by solving the Ramsey problem that if the central banks in these countries want to reduce the observed volatility of inflation and the output gap, they will need to increase their exchange rate intervention. We conclude that risk premium shocks do not produce a significant trade-off between inflation and growth. A central bank can substantially reduce the observed volatility of inflation and the output gap by changing the interest rate when the exchange rate is fluctuating in response to these shocks.

Third, whether there is a trade-off for the central bank after a real depreciation depends crucially on the mix of shocks hitting the economy. In a more complex environment with more shocks hitting the economy simultaneously (for example, if the changes in exports are offset by negative shocks to commodity prices), the overall result of a real depreciation could still be a contraction in GDP caused by the balance sheet effect. Therefore, the central bank should accentuate the output contraction if it wants to keep inflation under control. This last result is in line with the evidence that many central banks would like to postpone a currency depreciation so as to delay, first, the contraction caused by the balance sheet effect and, second, the additional contraction that the central banks need to enact in order to control the inflation rate.

The paper is organized as follows. Section 2 provides a detailed description of the model. In Section 3, we present the empirical strategy. The estimation results, parameters, variance decompositions, and impulse response functions are presented in Section 4. In Section 5 we implement the Ramsey approach. Section 6 discusses the caveats and scope of this paper. Finally, Section 7 concludes.

2. The structural model

Our model resembles others found in the recent literature, but we have adapted it to capture the essentials of small open economies and facilitate estimation. General references for this type of models include Woodford (2003), Clarida, Galí, and Gertler (1999, 2002), Galí and Monacelli (2005), and Galí, López-Salido, and Vallés (2007). More specifically, the model is similar to the one proposed by Smets and Wouters (2002). Our model also includes restricted consumers (Galí et al., 2007), raw materials, consumer habits, wage indexation, the balance sheet effect of exchange rate changes (Céspedes et al., 2004), and the country risk premium, which depends on the external debt–GDP ratio (Schmitt-Grohé & Uribe, 2003). Our structure is also similar to the one proposed by Laxton and Pesenti (2003), since all imports are intermediate inputs. The model thus incorporates imperfect pass-through of exchange rate changes to domestic prices. Previously, the model has been used for analyzing inflation-targeting regimes and fiscal policies in open economies (García, Restrepo, & Roger, 2011; García, Restrepo, & Tanner, 2011).

2.1. Households

We assume a continuum of infinitely lived households indexed by $i \in [0,1]$. As in Galí et al. (2007), a fraction of households, λ, consume their current labor income; they do not have access to capital markets and hence neither save nor borrow. Such agents have been called hand-to-mouth consumers. The remainder, 1–λ, have access to capital markets and are able to smooth consumption. Therefore, their intertemporal allocation between consumption and savings is optimal (that is, they are Ricardian or optimizing consumers).

2.1.1. Ricardian household consumption

The representative household maximizes expected utility,

$$E_o \sum_{t=0}^{\infty} \beta^t U\left(C_t^o(i), N_t^o(i)\right), \tag{1}$$

subject to the budget constraint

$$\begin{aligned} P_t C_t^o(i) \leq & W_t(i) N_t^o(i) + B_t^o(i) - S_t B_t^{o*}(i) + D_t^o(i) - P_t T_t \\ & - R_t^{-1} B_{t+1}^o(i) + S_t \left(\Phi\left(\frac{B_t^*}{GDP_t}, Q_t, u_t^{RP}\right) R_t^*, \right)^{-1} B_{t+1}^{o*}(i), \end{aligned} \tag{2}$$

where $C_t^o(i)$ is consumption, $D_t^o(i)$ are dividends from the ownership of firms, $\Phi\left(\frac{B_t^*}{GDP_t}, Q_t, u_t^{rp}\right)$ represents the country risk premium, S_t is the nominal exchange rate, $B_t^{o^*}(i)$ denotes private net foreign assets (where a positive value implies external debt), $W_t(i)$ is the nominal wage, $N_t^o(i)$ is the number of hours of work, $B_t^o(i)$ is government debt held by households, R_t and R_t^* are the gross nominal return on domestic and foreign assets, respectively (where $R_t = 1 + i_t$ and $R_t^* = 1 + i_t^*$), and T_t represents lump-sum taxes.

We have two alternative functional forms for the utility function: separable and the Greenwood–Hercowitz–Huffman (GHH) case. We include the GHH utility function because some authors claim that this function may replicate the volatility observed in small open economies better than the standard case (for example, Correia, Neves, & Rebelo, 1995). We also include the possibility of habit persistence to capture some lags in the response of consumption to different shocks. We write the two utility functions

$$U(C(i), N(i)) = \frac{\left(C_t(i) - \gamma C_{t-1}(i)\right)^{1-\sigma} - 1}{1-\sigma} - \psi \frac{N_t(i)^{\varphi}}{\varphi} \tag{3.1}$$

in the separable case and

$$U(C(i), N(i)) = \frac{\left(C_t(i) - \gamma C_{t-1}(i) - \psi N_t(i)^{\varphi}\right)^{1-\sigma} - 1}{1-\sigma} \tag{3.2}$$

in the GHH case, where 1/σ is the intertemporal elasticity of substitution in consumption and 1/(φ–1) is the elasticity of labor supply to wages in both cases. The value of ψ is calibrated to obtain a realistic fraction of steady-state hours worked. The first-order condition for consumption is

$$1 = \beta E_t \left(\Lambda_{t,t+1} R_t \left(\frac{P_t}{P_{t+1}} \right) \right) \tag{4}$$

where the stochastic discount factor is $\Lambda_{t,t+1}$. From the first-order conditions, it is also possible to derive the interest parity condition, where $Q_t^* = S_t/P_t$:

$$1 = E_t \left[\Lambda_{t,t+1} \frac{Q_{t+1}^*}{Q_t^*} \left(R_t^* \Phi\left(\frac{B_t^*}{GDP_t}, Q_t, u_t^{RP} \right) \right) \right] \tag{5}$$

Empirically, Eq. (5) is unable to generate a hump-shaped response of the real exchange rate after a shock to monetary policy (Adolfson, Laséen, Lindé, & Villani, 2008). We therefore assume that the real exchange rate, Q_t (Eq. (6)), is a weighted average between its own lag and the real exchange rate from the interest parity condition, Q_t^* (Eq. (5)):

$$Q_t = \left(Q_{t-1}\right)^{\Omega_Q} \left(Q_t^*\right)^{1-\Omega_Q} \tag{6}$$

2.1.2. Risk premium and risk premium shocks

As in Céspedes et al. (2004),[3] the risk premium, $\Phi\left(\frac{B_t^*}{GDP_t}, Q_t, u_t^{rp}\right)$, depends on debt, the exchange rate, GDP, and a risk premium shock, u_t^{rp}. The first term in the equation says that the risk premium is an increasing function of the ratio of external debt to GDP. This friction in the international capital markets is required to ensure stationarity of the external-debt-to-GDP ratio.[4] The second

[3] See also Gertler et al. (2007).

[4] See Schmitt-Grohé and Uribe (2003).

term captures the adverse impact of currency depreciation on the value of external debt in domestic currency—that is, the balance sheet effect. As the debt service burden on borrowers rises, the risk premium increases. We measure this effect for the parameter μ, which denotes the elasticity of the risk premium to the real exchange rate Q_t.[5] Finally, the third term is the risk premium shock, which we define as unanticipated changes in credit risk conditions related to external debt. As can be seen in the budget constraint in Eq. (1), this type of shock directly reduces the resources that families have available for smoothing consumption over time because they must pay a higher interest rate. In other words, in our study the shocks produce a credit spread between R_t^* and $R_t^*\Phi\left(\frac{B_t^*}{GDP_t}, Q_t, u_t^{rp}\right)$ that reduces households' resources.

With regard to the transmission mechanism, the risk premium shock affects both the real exchange rate and the risk premium simultaneously through Eq. (5), due to our general equilibrium approach. This occurs since the relevant foreign interest rate for the country is $R_t^*\Phi\left(\frac{B_t^*}{GDP_t}, Q_t, u_t^{rp}\right)$. Thus, an increase in this expression should produce a real depreciation caused by capital outflows. At the same time, there should be a second effect that reinforces the increase in the risk premium. In fact, the real depreciation raises the risk premium even further through the balance sheet effect, as captured by the second term in $\Phi\left(\frac{B_t^*}{GDP_t}, Q_t, u_t^{rp}\right)$. Nevertheless, the initial impulse in both variables was originally occasioned by a particular risk premium shock.

Although a risk premium shock clearly has a negative impact on the economy (over private consumption), this paper presents evidence that the economy must eventually respond to the shock with an expansion in the export sector and a corresponding increase in GDP. This result is the opposite of what many authors find when they introduce domestic financial shocks to explain the decline in GDP as a result of worse conditions in the credit market following the collapse of the financial system in 2007.[6] We can explain this apparent contradiction because in our study the risk premium shocks are not only financial, but also foreign in nature. Under certain circumstances, the economy could respond to these shocks with a real depreciation that allows it to stimulate the export sector, an element that is not present in models of large economies (see Section 6 for a discussion of caveats).

2.1.3. Hand-to-mouth household consumption

The utility of the credit-restricted households is given by

$$U\left(C_t^r(i), N_t^r(i)\right). \tag{7}$$

We assume that these households neither save nor borrow (Mankiw, 2000). As a result, their level of consumption is given by their disposable income:

$$P_t C_t^r(i) = W_t(i) N_t^r(i) - P_t T_t. \tag{8}$$

2.1.4. The labor supply schedule

Following Erceg, Henderson, and Levin (2000), we assume that households act as price setters in the labor market. There is a representative labor aggregator, and wages are staggered à la Calvo (1983). Therefore, wages can only be optimally changed after some random wage-change signal is received. A continuum of monopolistically competitive households supplies a differentiated labor service to the intermediate-good-producing sector. The representative labor aggregator combines household labor hours in the same amount that firms demand them, based on a constant-returns technology. The aggregate labor index, N_t, has the constant elasticity of substitution (CES) or Dixit–Stiglitz form, where ε_w is the elasticity of substitution between any two differentiated households (see Eq. (11) below):

$$N_t = \left[\int_0^1 N_t(i)^{\frac{\varepsilon_w - 1}{\varepsilon_w}} di\right]^{\frac{\varepsilon_w}{\varepsilon_w - 1}} \tag{9}$$

where $N_t\,(i)$ is the quantity of labor provided by each household.

The representative labor aggregator takes each household's wage rate, $W_t\,(i)$, as given and minimizes the cost of producing a given amount of the aggregate labor index. Units of labor index are then sold at their unit cost, W_t, to the productive sector (with no profit):

$$W_t = \left[\int_0^1 W_t(i)^{1-\varepsilon_w} di\right]^{\frac{1}{1-\varepsilon_w}} \tag{10}$$

Households set their nominal wages to maximize their intertemporal objective function (1), subject to the intertemporal budget constraint (2) and to the total demand for labor services, which is given by

$$N_t(i) = \left[\frac{W_t(i)}{W_t}\right]^{-\varepsilon_w} N_t \tag{11}$$

[5] See Céspedes et al. (2004) and Morón and Winkelried (2005).

[6] For instance: Gilchrist, Yankov, and Zakrajšek (2009), Kose, Otrok, Huidrom, and Helbling (2011), and, from a more historical perspective, Bordo, Haubrich, and J. (2010).

Additionally, we impose two important assumptions. First, rule-of-thumb households set their wages equal to the average wage of optimizing households. Second, Ricardian household consumers who do not receive the signal to change their nominal wage can index their wages to past inflation. We measure the level of indexation for δ_W. Thus, households that cannot reoptimize adjust their wages according to

$$W_t(i) = (W_{t-1}(i))^{1-\delta_W}\left(\frac{P_{t-1}}{P_{t-2}}\right)^{\delta_W} \tag{12}$$

2.1.5. Domestic intermediate-good firms

We assume a continuum of monopolistically competitive firms, indexed by $j \in [0\ \ 1]$ producing differentiated intermediate goods. The production function of the representative intermediate-good firm, indexed by (j) corresponds to a CES combination of labor, $N_t\,(j)$ and import inputs, $I_t\,(j)$, to produce $Y_t^D(j)$ and is given by

$$Y_t^D(j) = A_t\left[\alpha N_t(j)_t^{\frac{\sigma_S-1}{\sigma_S}} + (1-\alpha)I_t^{\frac{\sigma_S-1}{\sigma_S}}(j)\right]^{\frac{\sigma_S}{\sigma_S-1}} \tag{13}$$

where A_t is the technology shock, $\boldsymbol{\sigma_s}$ is the elasticity of substitution between capital and labor, and both are greater than zero.

The firms' costs are minimized, taking as given the price of import inputs, $S_tP_t^*$ and the wage, W_t, subject to the production function technology. The relative factor demands are derived from the first-order conditions:

$$\frac{S_tP_t^*}{W_t} = \left(\frac{\alpha}{1-\alpha}\right)\left(\frac{N_t(j)}{I_t^*(j)}\right)^{\frac{1}{\alpha_S}} \tag{14}$$

or

$$I_t^*(j) = \left(\frac{\alpha}{1-\alpha}\right)^{\frac{1}{\alpha_S}}\left(\frac{S_tP_t^*}{W_t}\right)^{-\frac{1}{\alpha_S}}N_t(j) \tag{15}$$

To replicate the inertia observed in the import process for inputs, we assume that total imports, I_t (Eq. (16)), are a weighted average between its own lag and the imports, I_t*, from (Eq. (15)):

$$I_t = (I_{t-1})^{\Omega_M}(I_t^*)^{1-\Omega_M} \tag{16}$$

The marginal cost is given by

$$MC^D = \frac{1}{A_t}\left[\alpha^{\sigma_s}(S_tP_t^*)^{1-\sigma_s} + (1-\alpha)^{\sigma_s}(W_t)^{1-\sigma_s}\right]^{\frac{1}{1-\sigma_S}} \tag{17}$$

When firm (j) receives a signal to optimally set a new price à la Calvo (1983), it maximizes the discounted value of its profits, conditional on the new price. Firms that do not receive a price signal index their prices to the last period's inflation, π_{t-1}, according to the parameter δ_D (that is, complete indexation is $\boldsymbol{\delta_D} = 1$). Therefore,

$$\max\sum_{k=0}^{\infty}\theta_D^kE_t\left\{\Lambda_{t,t+k}Y_{t+k}^D(j)\left(P_t^{D*}(j)\prod_{l=1}^k\left(\pi_{t+l-1}^k\right)^{\delta_D} - MC_{t+k}^D\right)\right\} \tag{18}$$

subject to

$$Y_{t+k}^D(j) \le \left(\frac{P_t^{D*}(j)}{P_t^D}\right)^{-\varepsilon_D}Y_{t+k}^D \tag{19}$$

where the probability that a given price can be reoptimized in any particular period is constant and is given by $(1–\theta_D)$, and ε_D is the elasticity of substitution between any two differentiated goods. $P_t^{D^*}$ must satisfy the first-order condition, where this price can be indexed to past inflation:

$$\sum_{k=0}^{\infty}\theta_D^kE_t\left\{\Lambda_{t,t+k}Y_{t+k}^D(j)\left(P_t^{D*}(j)\prod_{l=1}^k\left(\pi_{t+l-1}^k\right)^{\delta_P} - \frac{\varepsilon_D}{\varepsilon_D-1}MC_{t+k}^D\right)\right\} = 0 \tag{20}$$

Firms that did not receive the signal will not adjust their prices. Those that do reoptimize choose a common price, $P_t^{D^*}$. Finally, the dynamics of the domestic price index, P_t^D are described by the following equation:

$$P_t^D = \left[\theta_D\left(P_{t-1}^D\pi_{t-1}^{\delta_D}\right)^{1-\varepsilon_D} + (1-\theta_D)\left(P_t^{D*}\right)^{1-\varepsilon_D}\right]^{\frac{1}{1-\varepsilon_D}} \tag{21}$$

2.1.6. Final-good distribution

There is a perfectly competitive aggregator, which distributes the final good using a constant-return-to-scale technology:

$$Y_t^D = \left(\int_0^1 Y_t^D(j)^{\frac{\varepsilon_K - 1}{\varepsilon_K}} dj \right)^{\frac{\varepsilon_K}{\varepsilon_K - 1}} \tag{22}$$

where $Y_t^D(j)$ is the quantity of the intermediate good (domestic or imported) included in the bundle that minimizes the cost of any amount of output, Y_t. The aggregator sells the final good at its unit cost, P_t, with no profit:

$$P_t^D = \left(\int_0^1 P_t^D(j)^{1-\varepsilon_K} dj \right)^{\frac{1}{1-\varepsilon_K}} \tag{23}$$

where P_t is the aggregate price index. Finally, the demand for any good, $Y_t^D(j)$, depends on its price $P(j)$, which is taken as given, relative to the aggregate price level, P_t:

$$Y_t^D(j) = \left(\frac{P(j)}{P_t} \right)^{-\varepsilon_K} Y_t^D \tag{24}$$

2.2. Exports

Foreign countries' demand for domestic exports is modeled as follows. By assumption, the demand for each set of differentiated domestic goods depends on total consumption abroad, $C_t^{D^*}$, which we assume is an external shock, and on the home price of domestic goods relative to their price in the foreign country:

$$X_t^{D*} = \left[\left(\frac{P_t^D}{S_t P_t^{D*}} \right) \right]^{-t_D} C_t^{D*} \tag{25}$$

We assume that, in practice, exports,X_t^D, respond more slowly to real exchange rates and foreign demand than the export demand obtained from the model, $X_t^{D^*}$:

$$X_t^D = \left(X_{t-1}^D \right)^{\Omega} \left(X_t^{D*} \right)^{1-\Omega} \tag{26}$$

Because we are considering small economies' exports of natural resources (commodities), the total value from these products is $S_t P_t^{cu} Q\_c$, where P_t^{cu} denotes the international price of the commodities and Q_c is the constant quantity supplied. For simplicity, supply is assumed to be price invariant in the (short-run) business cycle horizon, and the commodity price P_t^{cu} is an external shock.

2.3. Aggregation

The weighted sum of consumption by Ricardian and rule-of-thumb agents yields an aggregate consumption of

$$C_t = \lambda C_t^r + (1-\lambda) C_t^o = \int_0^{\lambda} C_t^r(i) di + \int_{\lambda}^1 C_t^o(i) di \tag{27}$$

Since only Ricardian households hold assets, these are equal to

$$B_t = (1-\lambda)\left(B_t^o\right) \tag{28}$$

Foreign assets (or debt) include fiscal assets, $B_t^{G^*}$, and privately held assets, $B_t^{o^*}$:

$$B_t^* = B_t^{G*} + (1-\lambda) B_t^{o*} \tag{29}$$

Hours worked are given by a weighted average of the labor supplied by each type of consumer:

$$N_t = \lambda N_t^r + (1-\lambda) N_t^o \tag{30}$$

Finally, in equilibrium each type of consumer works the same number of hours:

$$N_t = N_t^r = N_t^o \tag{31}$$

2.4. Monetary policy

The central bank sets the nominal interest rate according to the following rule:

$$R_t = \bar{R}\left(\left(\frac{\prod_t}{\bar{\prod}}\right)^{\phi_\pi}\left(\frac{YR_t}{\overline{YR}_t}\right)^{\phi_y}\left(\frac{Q_t}{\bar{Q}}\right)^{\zeta_e^1}\left(\frac{Q_t}{Q_{t-1}}\right)^{\zeta_e^2}\right) \tag{32}$$

where $\bar{R}$ is the steady-state nominal interest rate, $\prod_t$ total inflation, $\bar{\Pi}$ total inflation in steady state (which is zero in our model), YR_t GDP excluding natural resources, $\bar{Y}\bar{R}$ the steady-state value, Q_t the real exchange rate, and $\bar{Q}$ the steady-state level. Thus, central banks can react to both the level of and the change in the real exchange rate.

We assume that central banks do not change immediately the interest rate to its target level (Eq. (32)); instead, they take some time to respond to changes in the inflation rate, output and the exchange rate (Eq. (33)), where u_t^{MP} are monetary policy shocks:

$$R_t = (R_{t-1})^{\Omega_R}(R_t^*)^{1-\Omega_R}\exp\left(u_t^{MP}\right) \tag{33}$$

2.5. Government

The government budget constraint is

$$P_tT_t + R_t^{-1}B_{t+1}^G + S_t\left(\Phi\frac{B_t^*}{GDP_t}, Q_t, u_t^{RP}\right)(R_t^*)^{-1}B_{t+1}^{G*} \geq B_t^G + S_tB_t^{G*} + P_t^GG_t, \tag{34}$$

where the country risk premium, $\Phi\left(\frac{B_t^*}{GDP_t}, Q_t, u_t^{rp}\right)$, is a positive function of foreign debt, B_t^G denotes public domestic assets (debt), P_t T_t corresponds to government nominal (lump-sum) tax revenues, and $P_t^GG_t$ is public spending. For simplicity, we assume that $G_t = 0$.

2.6. Market-clearing conditions

The market-clearing conditions for production factors are total employment by all firms j,

$$N_t = \int_0^1 N_t(j)dj \tag{35}$$

and import inputs,

$$I_t = \int_0^1 I_t(j)dj \tag{36}$$

In the goods market, the market-clearing condition is

$$Y_t^D = \left(C_t + X_t^D\right) \tag{37}$$

where the total supply of domestic goods equals the total demand for domestically produced goods (consumption plus exports).

Finally, the economy-wide budget constraint can be expressed as follows:

$$P_tC_t \leq P_t^DY_t^D - S_tP_t^*I_t + S_t\left(\Phi\left(\frac{B_t^*}{GDP_t}, Q_t, u_t^{RP}\right)R_t^*\right)^{-1}B_{t+1}^* - S_tB_t^* + (S_tP_t^{cu}Q\_c) \tag{38}$$

We define output excluding natural resources as the sum of domestically produced goods minus import inputs:

$$P_tYR_t = P_t^DY_t^D - S_tP_t^*I_t \tag{39}$$

3. Econometric methodology: A VAR prior from the general equilibrium model

To measure monetary policy, we apply the strategy proposed by Del Negro, Schorfheide, Smets, and Wouters (2007). Specifically, we use the model described in the last section to obtain the prior information for estimating a vector autoregression (VAR) model. First, we use a Bayesian approach to estimate the DSGE model: we define a prior distribution for the vector of parameters θ of the DSGE model and then use these priors to obtain the priors for the VAR model (namely, the vector of parameters ϕ and the covariance matrix Σ_u). These new priors are denoted $\phi(\theta)$ and $\Sigma_u(\theta)$, but we allow deviations from the restrictions imposed by the DSGE in order to capture potential misspecification. Thus, the accuracy of the prior is measured by a hyperparameter, λ_{DSGE}. This creates a continuum of models, that Del Negro et al. (2007) term a DSGE-VAR. They show that when hyperparameter λ_{DSGE} is close to zero, the model converges to an unrestricted VAR; and when hyperparameter λ_{DSGE} tends to infinity, the model converges to the DSGE model.

In this approach, the parameter λ_{DSGE} is estimated by achieving the highest marginal density. By construction, this estimation attains a better fit and tends to deliver more reliable impulse responses than the restricted model (that is, the DSGE model). This approach aims to keep the sequence of auto-covariance associated with the DSGE-VAR as close as possible to the DSGE model without sacrificing the DSGE-VAR model's ability to match historical data.

We can also use the DSGE-VAR as a benchmark for evaluating our dynamic general equilibrium models. Thus, strong deviations of the DSGE-VAR parameters relative to the restrictions imposed by the DSGE model are construed as misspecification problems in our DSGE model.[7]

4. Priors and results

Table 1 lists the values of the priors. These values are in line with the earlier literature and incorporate beliefs about possible ranges based on the nature and behavior of the variables (see Laxton & Pesenti, 2003; Smets & Wouters, 2002). One of the advantages of the Bayesian method is that it gives a voice to the data, supplying information about how well the parameters fit the data and the economic reality. The values of the parameters used in DSGE models in the different countries fall within the literature's typical range. Accordingly, we use the same prior values for the countries in our sample, thereby letting the data reveal the degree of fit of these values to the realities of the sample countries.

4.1. Utility functions

As mentioned earlier, we used two different types of utility functions to estimate the model—separable and GHH. Surprisingly, the separable utility function gave the best results in all countries except Colombia (see Table 2 with posterior odds). This may be explained by the presence of heterogeneous (hand-to-mouth) agents, which replicates the high volatility observed in these economies better than a GHH utility function. This is an important result since some authors propose that the GHH utility function better captures the higher volatility observed in small open economies (Correia et al., 1995). Our results indicate that the presence of restricted agents and a low degree of habit should provide a better alternative to produce this volatility in some countries.

The parameter estimates of the impact of monetary policy on the economies are presented in Table 3. A first important result is that the estimation of σ for all countries is around 2.0. This implies an intertemporal substitution elasticity of 0.5, which confirms that the interest rate has a moderate effect on consumption in emerging economies (Agénor & Montiel, 1996).

Another parameter that is related to the response of consumption to the interest rate is the habit parameter, γ. Our estimations indicate that the presence of habit is moderate (10–20 percent) relative to developed economies (Christiano, Eichenbaum, & Evans, 2005). We also estimate the effect of restricted agents on aggregate consumption, λ. These agents account for around 19 to 29 percent of total consumption in our sample of countries.

Prices, remain rigid, on average, calculated by $1/(1–\theta_D)$, between three and five quarters. Importantly, the rigidity of prices and wages, $1/(1–\theta_W)$, tends to be quite similar for all countries under study. Another important result is that the level of indexation is between 40 and 50 percent and again is similar to prices, δ_D, and wages, δ_W. This indicates that there is some degree of connection in the setting of prices and wages in these economies, which produces important real rigidities in the labor market. Furthermore, in the model all imported goods are inputs for production, so price rigidity also indicates a low pass-through of the exchange rate to domestic prices.

Other result relevant for understanding monetary transmission channel is the elasticity of differentiated goods exports to the real exchange rate, τ_D. The estimated value is between 1.5 and 2.0. This result, together with price rigidity, indicates that reallocations of the real exchange rate in these economies are significant (Colacelli, 2008). In contrast, we find that the inertia of domestic exports, Ω, and imports of inputs, Ω_M, are below 0.2, reaffirming the strong impact of the real exchange rate on the economy in the short run.

The balance sheet effect may be positive or negative, depending on the structure of each economy. Nevertheless, we force our prior for the parameter μ to be positive for Peru, given the evidence found by Morón and Winkelried (2005). As explained earlier, in our model, this effect is captured arbitrarily by incorporating the real exchange rate equation in the risk premium, $\Phi\left(\frac{B_t^*}{GDP_t}, Q_t, u_t^{rp}\right)$. In all cases, the parameter μ, which captures the elasticity of the risk premium to the real exchange rate, Q_t, turns out to be positive. New Zealand and Australia has a μ below 0.3, with the other countries ranging up to 0.5.

Another important theoretical relationship to be tested is the uncovered interest parity condition. Our results indicate that this parity does not hold as expected in any of the countries. The persistence of the real exchange rate, Ω_Q, is around 0.3, while for Peru it is 0.6. This is half the value obtained by Adolfson et al. (2008) in a DSGE model estimated with Bayesian econometrics for the case of Sweden.

On the Taylor rule, we find that persistence (Ω_R), inflation (ϕ_Π) and output (ϕ_Y) are around 0.7, 2.0, and 0.5, respectively. In this sense, the Taylor rule is very similar to that found in other economies (Woodford, 2003). The fundamental difference is that central banks in these emerging economies also respond moderately to the real exchange rate—in terms of both the level, ζ^1_e, and the volatility, ζ^2_e. This result is repeated systematically in all countries and in contrast to other studies, such as Lubik and Schorfheide (2007).

[7] A complete description of data and method of solution is provided in Appendix A.

Table 1
Prior distribution for small open economies.

Parameters	Distribution	Prior	
		Mean	s.e.
σ	gamm	2.00	0.10
Ω_M	gamm	0.25	0.05
T_D	gamm	5.00	0.75
Ω	beta	0.20	0.05
γ	beta	0.10	0.05
λ	gamm	0.35	0.05
Ω_R	beta	0.50	0.05
φ_Π	gamm	2.00	0.10
φ_Y	gamm	0.50	0.10
ζ_e^1	gamm	0.25	0.10
ζ_e^2	gamm	0.25	0.10
$\mu(^*)$	unif	0.00	0.50
Ω_D	beta	0.60	0.10
θ_D	beta	0.65	0.05
δ_D	beta	0.45	0.05
θ_W	beta	0.65	0.05
δ_W	beta	0.45	0.05
λ_{DSGE}	unif	3.00	2.00
Autocorrelation coef.			
ρ_Y Foreign output	beta	0.50	0.10
ρ_P Foreign inflation	beta	0.50	0.10
ρ_R Foreign interest rate	beta	0.50	0.10
ρ_A Productivity	beta	0.50	0.10
ρ_{PC} Price commodity	beta	0.50	0.10
ρ_{Z2} Preference	beta	0.50	0.10
ρ_{Z3} Mark-up prices	beta	0.50	0.10
ρ_{Z4} Risk premium	beta	0.50	0.10
ρ_{Z5} Mark-up wages	beta	0.50	0.10
ρ_{MA1}	beta	0.50	0.10
ρ_{MA2}	beta	0.50	0.10
ρ_{MA3}	beta	0.50	0.10
ρ_{MA4}	beta	0.50	0.10
Std. coef.			
σ_M Interest rate	invg	0.50	0.60
σ_Y Foreign output	invg	2.00	2.00
σ_P Foreign inflation	invg	2.00	2.00
σ_R Foreign interest rate	invg	2.00	2.00
σ_A Productivity	invg	2.00	2.00
σ_{PC} Price commodity	invg	8.00	4.00
σ_{Z2} Preference	invg	2.00	2.00
σ_{Z3} Mark-up prices	invg	2.00	2.00
σ_{Z4} Risk premium	invg	2.00	2.00
σ_{Z5} Mark-up wages	invg	2.00	2.00

4.2. Variance decomposition: The role of risk premium shocks and commodity price volatility in the business cycle of small open economies

The main result that emerges from the variance decomposition n periods ahead is that, in addition to the standard shocks studied in developed economies, we need to consider the risk premium shock to explain macroeconomic variables in small open economies (see Fig. 1 in Appendix B). This shock largely explains the variability of the real exchange rate in conjunction with the external interest rate. It also explains a lower percentage of GDP fluctuations and, to a lesser extent, the behavior of the nominal interest rate. This shock does not explain the variability of inflation, but it is the most significant external shock for exchange

Table 2
Posterior odds for alternative models.

	GHH	Separable	GHH DSGE-VAR	Separable DSGE-VAR
Australia	−767.67	−681.94	−685.29	−613.19
New Zealand	−816.78	−725.63	−755.16	−688.36
Chile	−845.67	−790.70	−781.28	−713.66
Colombia	−990.44	−943.14	−892.72	−900.37
Peru	−793.97	−790.14	−796.34	−721.35

Table 3
Posterior distribution for emerging economies.

Parameters	Australia			New Zealand			Chile			Colombia			Peru		
	Separable DSGE-VAR			Separable DSGE-VAR			Separable DSGE-VAR			GHH DSGE-VAR			Separable DSGE-VAR		
	Mean	5%	95%	Mean	5%	95%	Mean	5%	95%	Mean	5%	95%	Mean	5%	95%
σ	1.97	1.84	2.13	2.01	1.88	2.13	2.01	1.88	2.18	2.00	1.84	2.15	1.99	1.82	2.14
Ω_M	0.20	0.14	0.25	0.17	0.13	0.22	0.22	0.18	0.26	0.14	0.09	0.18	0.19	0.13	0.25
T_D	1.93	1.60	2.23	1.76	1.59	1.95	2.10	1.61	2.49	1.95	1.59	2.41	3.62	2.82	4.40
Ω	0.25	0.20	0.30	0.23	0.16	0.30	0.25	0.18	0.32	0.42	0.33	0.52	0.21	0.14	0.28
γ	0.10	0.04	0.17	0.08	0.02	0.17	0.11	0.05	0.17	0.17	0.06	0.28	0.09	0.03	0.16
λ	0.21	0.16	0.25	0.19	0.14	0.23	0.22	0.16	0.28	0.29	0.24	0.35	0.21	0.16	0.25
Ω_R	0.74	0.69	0.79	0.66	0.60	0.72	0.72	0.67	0.78	0.44	0.37	0.50	0.42	0.36	0.49
ϕ_Π	1.88	1.75	2.00	1.87	1.72	2.00	1.96	1.81	2.11	2.05	1.92	2.19	2.00	1.84	2.17
ϕ_Y	0.54	0.39	0.69	0.66	0.51	0.81	0.39	0.27	0.51	0.83	0.68	0.97	0.51	0.33	0.69
ζ_e^1	0.10	0.05	0.15	0.13	0.06	0.19	0.12	0.05	0.19	0.13	0.06	0.20	0.42	0.18	0.66
ζ_e^2	0.12	0.04	0.19	0.17	0.10	0.24	0.22	0.10	0.32	0.13	0.06	0.20	0.43	0.25	0.61
$\mu(^*)$	0.28	0.16	0.40	0.22	0.13	0.32	0.30	0.20	0.40	0.54	0.39	0.69	0.36	0.33	0.40
Ω_D	0.27	0.21	0.34	0.26	0.18	0.34	0.31	0.23	0.39	0.29	0.20	0.38	0.54	0.35	0.74
θ_D	0.68	0.61	0.76	0.70	0.65	0.76	0.75	0.70	0.79	0.74	0.68	0.80	0.79	0.74	0.84
δ_D	0.39	0.33	0.45	0.41	0.36	0.45	0.43	0.36	0.48	0.43	0.37	0.49	0.47	0.40	0.55
θ_W	0.65	0.58	0.71	0.64	0.58	0.68	0.66	0.60	0.73	0.68	0.63	0.72	0.67	0.60	0.74
δ_W	0.47	0.42	0.53	0.44	0.37	0.50	0.48	0.42	0.53	0.46	0.40	0.55	0.44	0.37	0.51
λ_{DSGE}	5.35	4.46	6.43	5.44	4.46	6.46	5.56	4.75	6.46	3.20	2.52	3.84	5.78	5.01	6.46
Autocorrelation coef.															
ρ_Y Foreign output	0.61	0.50	0.74	0.66	0.56	0.76	0.61	0.47	0.75	0.53	0.36	0.69	0.59	0.43	0.75
ρ_P Foreign inflation	0.55	0.40	0.71	0.65	0.54	0.79	0.60	0.49	0.72	0.65	0.54	0.77	0.62	0.47	0.78
ρ_R Foreign interest rate	0.65	0.56	0.74	0.62	0.51	0.73	0.59	0.45	0.73	0.66	0.57	0.78	0.64	0.51	0.77
ρ_A Productivity	0.29	0.16	0.40	0.34	0.19	0.53	0.38	0.28	0.48	0.39	0.28	0.51	0.41	0.26	0.54
ρ_{PC} Price commodity	0.55	0.42	0.68	0.65	0.54	0.77	0.51	0.39	0.62	0.45	0.26	0.64	0.53	0.38	0.68
ρ_{Z2} Preference	0.48	0.34	0.62	0.55	0.41	0.67	0.53	0.41	0.68	0.49	0.40	0.58	0.48	0.34	0.60
ρ_{Z3} Mark-up prices	0.44	0.25	0.60	0.49	0.40	0.58	0.31	0.20	0.43	0.35	0.24	0.45	0.41	0.27	0.54
ρ_{Z4} Risk premium	0.43	0.28	0.56	0.43	0.31	0.55	0.46	0.37	0.56	0.48	0.34	0.61	0.39	0.27	0.52
ρ_{Z5} Mark-up wages	0.60	0.50	0.70	0.52	0.36	0.63	0.54	0.40	0.67	0.47	0.27	0.65	0.48	0.31	0.65
ρ_{MA1}	0.39	0.26	0.50	0.33	0.25	0.41	0.43	0.28	0.59	0.34	0.23	0.44	0.36	0.23	0.50
ρ_{MA2}	0.54	0.45	0.63	0.43	0.33	0.53	0.42	0.31	0.51	0.43	0.32	0.53	0.47	0.32	0.60
ρ_{MA3}	0.32	0.20	0.43	0.46	0.33	0.57	0.41	0.27	0.59	0.40	0.25	0.56	0.42	0.27	0.56
ρ_{MA4}	0.50	0.38	0.62	0.50	0.39	0.58	0.59	0.44	0.72	0.48	0.35	0.60	0.52	0.33	0.71
Std. coef.															
σ_M Interest rate	0.22	0.15	0.29	0.35	0.24	0.44	0.36	0.24	0.47	0.67	0.50	0.85	1.05	0.80	1.30
σ_Y Foreign output	0.57	0.46	0.67	0.59	0.48	0.70	0.63	0.50	0.75	0.57	0.46	0.68	0.61	0.49	0.73
σ_P Foreign inflation	0.41	0.33	0.47	0.42	0.33	0.49	0.39	0.33	0.46	0.37	0.31	0.43	0.40	0.33	0.47
σ_R Foreign interest rate	0.37	0.31	0.41	0.37	0.31	0.42	0.37	0.31	0.42	0.37	0.32	0.42	0.38	0.31	0.44
σ_A Productivity	2.56	1.92	3.15	3.12	2.35	3.80	3.45	2.66	4.14	2.44	1.82	3.00	6.15	4.64	7.60
σ_{PC} Price commodity	3.24	2.68	3.84	4.12	3.37	4.87	8.63	7.11	10.25	8.30	6.52	9.96	9.21	7.38	11.04
σ_{Z2} Preference	0.56	0.42	0.69	0.60	0.46	0.75	0.83	0.55	1.08	1.86	1.24	2.41	0.88	0.62	1.14
σ_{Z3} Mark-up prices	0.54	0.40	0.68	0.49	0.38	0.60	0.55	0.42	0.69	0.51	0.39	0.62	0.50	0.38	0.62
σ_{Z4} Risk premium	0.81	0.59	1.06	0.95	0.68	1.22	0.72	0.54	0.91	1.08	0.77	1.38	1.16	0.85	1.46
σ_{Z5} Mark-up wages	0.43	0.35	0.51	0.47	0.38	0.56	0.40	0.32	0.47	0.59	0.48	0.70	0.62	0.49	0.74

rate.[8] In contrast, the commodity price shock is only relevant in explaining the volatility of GDP in Chile and Peru. The external GDP shock and the external inflation shock do not appear to be relevant in the period considered.

Other shocks that have been used in the literature to explain the fluctuations also appear in our results. Preference shocks are important in explaining fluctuations in GDP; mark-up shocks are a decisive factor for the variability of inflation and interest rates; and productivity shocks are important in explaining all the variables. In contrast, monetary shocks are present in all the variables, but their relevance is small. This does not mean that monetary policy is ineffective; on the contrary, it works through the response to the other shocks to stabilize the economy.[9]

4.3. Impulse responses: Monetary policy in the face of risk premium and commodity price shocks

A risk premium shock generates a sharp increase of the real exchange rate. This causes the central banks to react strongly by increasing the monetary policy interest rate (see Fig. 2 in Appendix B). The increase in the real exchange rate stimulates exports and growth, which in turn pushes up inflation. In this scenario, raising the monetary policy rate is not contradictory with the goals

[8] This evidence is consistent with Chen, Rogoff, and Rossi (2010), who find that the exchange rate cannot be projected based on commodity prices, although commodity prices can be forecast using the exchange rate.

[9] In fact, monetary policy has a standard effect on the economy (see Fig. 3 in Appendix B).

of reducing inflation, stabilizing growth, and smoothing the real exchange rate. It is therefore convenient for central banks to respond to fluctuations in the real exchange rate: raising the interest rate will reduce inflation, output, and the real exchange rate simultaneously, without any trade-off as in the case of a mark-up shock.

With regard to the commodity price shock (Fig. 3 in Appendix B), an increase in this variable expands output growth in some countries. It has a small effect on inflation, but it tends to appreciate the real exchange rate. The monetary policy response to this shock is a slight reduction in the interest rate as a result of the exchange rate appreciation. Interestingly, the central banks do not respond in the standard way (that is, by increasing the interest rate when output rises). Instead, they prefer to avoid further appreciation of the exchange rate by implementing small reductions in the policy rate.

Our results for the other shocks—namely, monetary (Fig. 4), productivity, mark-up, and preference shocks—are in line with the discussion in the literature. See García and González (2009) for a more detailed discussion of the effects of these shocks on these emerging economies.

4.4. Discussion of the results

The expansionary effect on output of a risk premium shock is controversial. In the traditional Mundell–Fleming–Dornbusch model, which is based on sticky good prices and a rapidly clearing asset market, the devaluation of the domestic currency is an essential element in the mechanism to adjust the economy following a negative external shock. After the Asian crisis, however, many authors proposed that a depreciation may be contractionary in the presence of foreign currency debt, an effect known as the balance sheet effect or the original sin.[10]

Our model also considers a type of balance sheet effect that acts through the impact of the exchange rate on the risk premium; this strategy is similar to Gertler, Gilchrist, and Natalucci (2007). Specifically, in our model, the parameter μ, which is the elasticity of the risk premium to the real exchange rate Q_t, measures the financial impact of increases in the interest rate on foreign currency debt caused by higher exchange rates. Despite the incorporation of this element in the model, the results indicate that a positive risk premium is still expansive.

To address the above point, Figs. 5 and 6 in Appendix B show the transmission of a risk-premium shock in the DSGE model under two alternative cases: the estimated case (the benchmark) and a counterfactual case (the balance sheet effect) in which we have held the parameter μ around 0.3, reduced the elasticity of exports to the exchange rate (one-third of the estimated parameter), and reduced the substitution elasticity between labor and imported inputs (from 0.75 to 0.4). This counterfactual scenario is favorable for obtaining the contractionary effect predicted by the balance sheet effect for a real devaluation of the domestic currency.

As the figures show, the contractionary effect is present only if the model parameters are very different from our estimation. In other words, the other effects considered in the model are stronger than the balance sheet effect. Recall that there are three additional channels through which the exchange rate can affect the economy in the model: the cost of imported inputs, natural resource exports, and intermediate good exports. Our model estimation indicates that the impact of a real depreciation on exports far exceeds not only the balance sheet effect, but also the effect of an increase in the cost of imported inputs. This last outcome could produce a *J* curve in the economy when exchange rate rises in the short run (Bahmani-Oskooee & Ratha, 2004).

Since our interpretation of the results depends critically on the impulse response functions, we checked whether these functions rely on a specific order of the endogenous variables, using different alternatives. Fig. 7 in Appendix B shows that for different orders, the estimated impulse response functions do not change significantly. In other words, we can confirm that the risk premium shock is expansive for the countries and the sample considered in this study.

5. The welfare implications of responding to a risk premium shock: The Ramsey approach

In this section, we compare the optimal response to a risk premium shock and the estimated response of the model. To do so, we calculate the Ramsey problem following the approach of Adão, Correia, and Teles (2004), Khan, King, and Wolman (2002) and Schmitt-Grohé and Uribe (2004) for models with sticky prices. For simplicity, we assume that the loss objective function is arbitrary, since our target is basically different from the objectives in these studies. We do not want to characterize optimal monetary policy given a specified utility function. Instead, we address the more empirical question of how central banks can improve their reaction to a risk premium shock given any objective function they choose. Accordingly, our loss function is defined in terms of the variance of inflation, the output gap, and the monetary policy rate:

$$Loss\quad function = \mathrm{var}(\Pi_t) + \varphi(\mathrm{var}(GDP)) + 0.2(Var(R_t)) \tag{40}$$

We follow the same strategy as Laxton and Pesenti (2003), and the parameter ϕ takes different values (namely, 0.5, 1.0, and 1.5).

As expected, there are important differences between the Ramsey allocations and the estimated response of the model (Fig. 8 in Appendix B). One possible explanation of this result is that there is no coincidence between our arbitrary loss function and the welfare aggregate function that can be obtained directly from the utility function and the rest of the equations of the model.

[10] Krugman (1999); Aghion, Bacchetta, and Banerjee (2004); Céspedes et al. (2004); Eichengreen and Hausmann (1999). Bahmani-Oskooee and Miteza (2003) assess the empirical evidence on whether devaluations are expansionary or contractionary. Empirical evidence on the balance sheet effect can be found in Berganza, Chang, and Herrero (2004) and Galindo, Panizza, and Schiantarelli (2003).

We could also argue that the central banks are not fully considering the welfare function to implement their monetary policies. In fact, some central banks are limited by law in terms of their targets, which may not be completely in line with the welfare criterion used in economic theory. Even if the central banks are actually using that optimal criterion to get their monetary policy rule, their utility function could still differ from the utility function that we have defined in our model.

Nevertheless, if the central banks decide to be concerned about the volatility of inflation and the output gap as established in Eq. (40), they can approximate to the Ramsey allocations by increasing the weight of the real exchange rate in the Taylor rule. In Fig. 8 in Appendix B we have arbitrarily increased the weight of the real exchange rate in the Taylor rule to 2.0 for the different values of φ. The results show that if one of the central banks wants to reduce the observed volatility of inflation and the output gap, it should be more active in the exchange rate market. Thus, in practice, to increase stabilization following Eq. (40), these emerging economies would have to intervene more strongly in the exchange rate market in order to reduce the volatility of shocks that directly affect this variable (that is, the exchange rate).

Defining a central bank's loss function is a controversial issue, but it facilitates the design of monetary policy once there is agreement about the bank's priorities and targets. For instance, Caballero and Lorenzoni (2007) consider intervention to be optimal if agents are financially constrained, one of the elements of our own model as well. Taking the Caballero–Lorenzoni model as representative of the economy, we show in our model that a central bank could implement that policy in practice by explicitly including the exchange rate in the Taylor rule with an appropriate coefficient.

5.1. Potential balance-sheet effects and trade-off for the central bank

Notwithstanding the previous result that after a real depreciation the balance sheet effect is weaker than the expansive response of exports of final products (e.g., industrial goods or services), there may still be a potential balance sheet effect if the depreciation occurs simultaneously with negative shocks. For example, Céspedes et al. (2004) illustrates the balance sheet effect in a setting with constant exports. One way of interpreting this assumption is that the expansionary effect of a real depreciation on exports of final products may be offset by other negative shocks, such as a negative commodity shock. Regarding this hypothesis, we confirm that during the Asian crisis, for which many authors have argued the presence of a balance sheet effect,[11] all countries that we consider in this study faced important negative shocks in commodity prices. Thus, given the offsetting effects that take place in the trade sector, the overall result could be a contraction of GDP deriving from the balance sheet effect that occurs in the financial sector. For example, Fig. 9 in Appendix B shows what happens if two shocks occur simultaneously in the model—a risk premium shock, which can be considered a pure exchange rate shock, and a negative shock to the commodity price. As the figure shows, the latter shock can entirely offset the expansionary effect of the real depreciation on the external demand for final goods that we found in the previous section, staying only the GDP contraction from the balance sheet effect.

The central bank will face an important trade-off if the effect of a real depreciation is contractionary. If we keep the assumption that the central bank's loss function is a weighted sum of the variance of inflation and output (see Eq. (40)), and then increasing interest rates to alleviate the effect on inflation, which is an optimal decision (Galí, 2008), will sharpen the economic contraction and, therefore, enlarge the variance of output in the loss function. Consequently, in practice, the central bank would like to postpone the currency depreciation to delay, first, the contraction of the balance sheet effect and, second, the additional contraction that the central bank needs to enact in order to control the inflation rate.

Therefore, whether the central bank faces a trade-off in response to a real depreciation depends crucially on the mix of shocks hitting the economy. In the case of a pure shock in the exchange rate (i.e., a risk premium shock), the central bank can easily stabilize the economy with changes in the interest rate. In a more complex environment, where the changes in exports are offset by negative shocks to commodity prices, the central bank should accentuate the contraction of the product if it wants to keep inflation under control.

6. Caveats

A more proactive exchange rate intervention has important costs in practice. For example, as suggested by Lahiri and Végh (2003), increasing the interest rate to defend the peg tends to delay the crisis, but higher interest rates increase the public debt service and imply higher future inflation, which tends to move the crisis forward. On the other hand, Morón and Winkelried (2005) point out that a more active intervention policy might increase the vulnerability of these economies by making them more dollarized.

The policy recommendations in this paper depend on the economic conditions in each country. However, as the economies have been able to normalize both their fiscal imbalances and their excess private foreign debt, a limited exchange rate stabilization may be helpful for reducing their volatility in the short term. This could be valid for small open economies that have reasonable levels of both foreign debt and inflation, as do most of the countries considered in this study. In these cases, the problems for policymakers are nominal rigidities and financial market imperfections, which produce suboptimal fluctuations in response to different shocks.

7. Conclusion

We have estimated a DSGE model for an emerging economy using Bayesian econometrics. Our results indicate that emerging economies face more challenges in designing monetary policy than developed economies. For instance, we find that the risk

[11] See for example, Aghion et al. (2004) Krugman (1999), Chang and Velasco (2000), Caballero and Krishnamurthy (2001).

premium shock could explain most of the variability in the real exchange rate. The results also indicate that the real exchange rate causes significant reallocation of resources across sectors in the short run.

One of the most important findings of our study is the expansionary effect on output of a risk premium shock. As we discuss in the text, this result is controversial. One conclusion in the Mundell–Fleming–Dornbusch model is that the devaluation of the domestic currency is an essential element in the mechanism to adjust the economy following a negative external shock. Nevertheless, this is at odds with the so-called balance sheet effect or original sin. This effect is present in our model, but the other effects are stronger. Our model estimation indicates that the impact of domestic currency depreciations on exports far exceeds not only the balance sheet effect, but also the effect of an increase in the cost of imported inputs, which could cause a *J* curve in the short run.

The paper shows that the monetary policy response is a sharp increase in the interest rate. This happens because the shock increases the real exchange rate, which stimulates exports and thus growth. As a result, the inflation rate also increases. In this scenario, the central bank is not faced with a trade-off between inflation and output because both variables are increasing simultaneously. Therefore, the central bank could respond quickly to this volatility by increasing the interest rate in order to stabilize both variables.

For instance, using the allocations of the Ramsey problem as a benchmark, this paper shows that, for the sample of selected countries, the central bank can reduce the observed volatility of inflation and the output gap with a more decisive intervention in the exchange rate market in order to reduce the volatility produced by the risk premium shocks. For instance, the central banks in these economies may approximate the Ramsey allocations by increasing the weight of the real exchange rate level in the Taylor rule to 2.0.

Nevertheless, the central bank will still face an important trade-off in a more complex environment with more shocks hitting the economy simultaneously, for example if the changes in exports are offset by negative shocks in commodity prices. We show that given the offsetting effects that take place in the trade sector, the overall result of a real depreciation could be a contraction of GDP caused by the balance sheet effect in the financial sector. Therefore, the central bank should accentuate the output contraction if it wants to keep inflation under control. Whether there is a trade-off for the central bank after a real depreciation depends crucially on the mix of shocks hitting the economy.

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Appendix A. Description of the data and method of solution

We use quarterly data ranging from 1994 to 2007. The variables observed are real GDP, real consumption, inflation, the nominal interest rate, the real exchange rate, and commodity prices. Commodity prices are measured in real terms, and the following commodities are used: for Chile and Peru, copper; for Colombia, WTI oil; for Australia, the commodity price index published by the Reserve Bank of Australia; for New Zealand, the soft commodity price index published by the Reserve Bank of New Zealand. Our data sources for Chile, Colombia, and Peru are the respective central banks, with the exception of the real exchange rate index, which is published by J.P. Morgan, and commodity prices, which are from Bloomberg. For Australia and New Zealand, all the data, except for the real exchange rate, are from the respective central banks.

External variables are from the Federal Reserve Economic Data (FRED) database, maintained by the Federal Reserve Bank of St. Louis. We observe the following variables: real GDP, the GDP deflator as a measure of inflation, and the interest rate (which corresponds to the Federal funds rate). The equations for measuring variables are given by

$$Y_t = \begin{bmatrix} Y_{obs} \\ C_{obs} \\ W_{obs} \\ Q_{obs} \\ PCM_{obs} \\ \Pi_{obs} \\ R_{obs} \\ Y^*_{obs} \\ R^*_{obs} \\ \Pi^*_{obs} \end{bmatrix} = \begin{bmatrix} \bar{\gamma} \\ \bar{\gamma} \\ \bar{\gamma} \\ \bar{\gamma} \\ \bar{\gamma} \\ \bar{\pi} \\ \bar{r} \\ \bar{\gamma}^* \\ \bar{r}^* \\ \bar{\pi}^* \end{bmatrix} + \begin{bmatrix} y_t - y_{t-1} \\ c_t - c_{t-1} \\ w_t - w_{t-1} \\ q_t - q_{t-1} \\ pcm_t - pcm_{t-1} \\ \pi_t \\ r_t \\ y^*_t - y^*_{t-1} \\ r^*_t \\ \pi^*_t \end{bmatrix}.$$

We detrend the model with a deterministic trend, Ξ, which was also estimated. The procedure was implemented for the example in this way: $c_t = C_t/\Xi^t$, $y_t = Y_t/\Xi^t$, and so on (Smets & Wouters, 2007). The model was the log-linearized around a nonstochastic steady state. The estimates, impulse responses, and variance decompositions were obtained with Dynare.[12] In our study, we followed the econometric methodology proposed by Del Negro and Schorfheide (2004), but with the improvements proposed by Adjemian, Darracq-Pariès, and Moyen (2008) for increasing the efficiency of the calculations through a direct estimation of the parameter λ_{DSGE}.

Therefore, given the observed variables, we need ten shocks to estimate the model. In Section 2, we explicitly define five shocks: productivity, monetary, price commodity, risk premium, and foreign demand. We then add five more shocks: a preference shock in the Euler equation, a markup shock in the Phillips curve, a wage shock, a foreign inflation shock, and a Federal funds shock.

[12] All this information (code and steady state) is available on request.

Appendix B. Figures

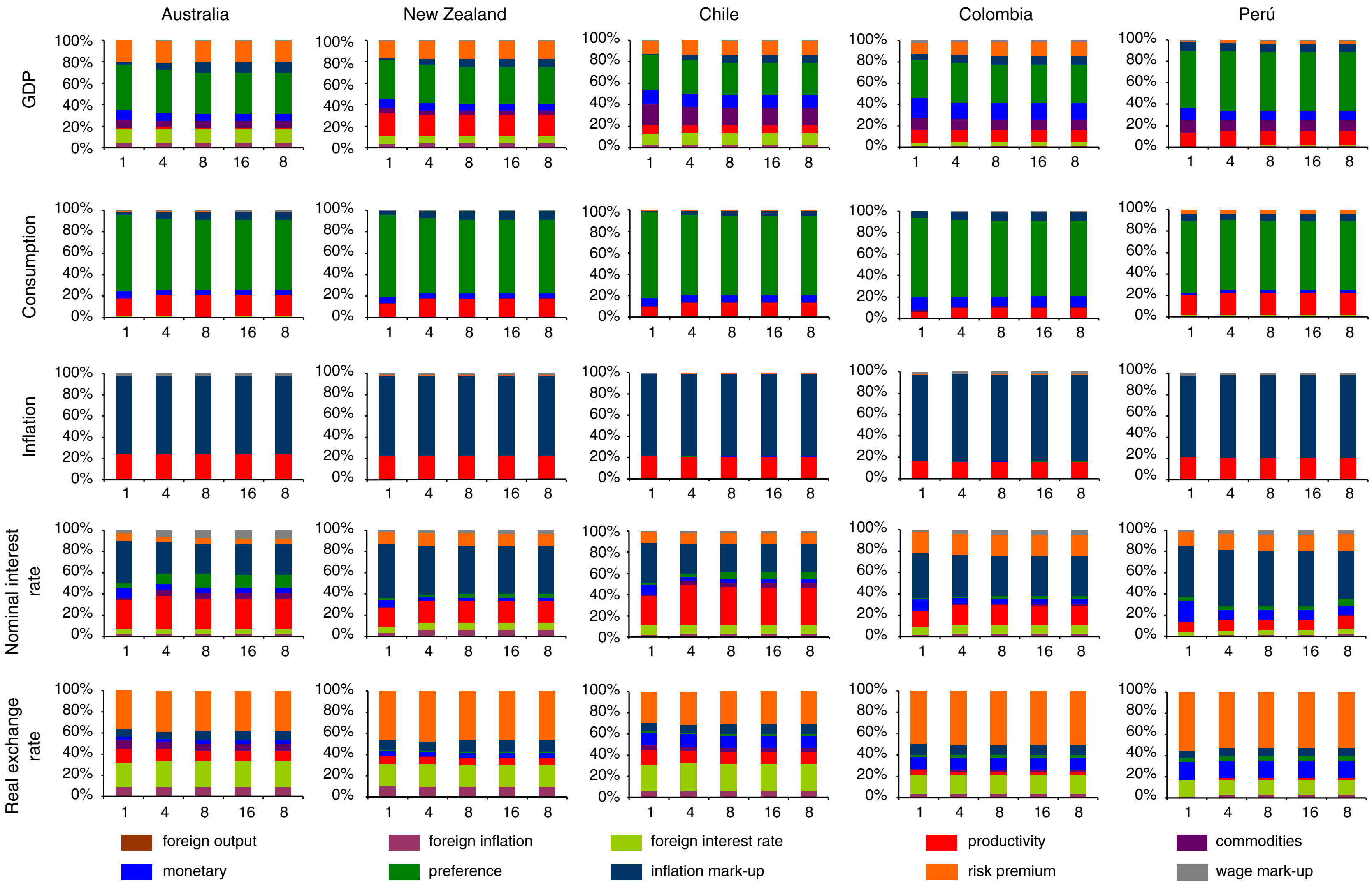

Fig. 1. DSGE-VAR variance decomposition. [a]
a. Computations conditional on the posterior mode values.

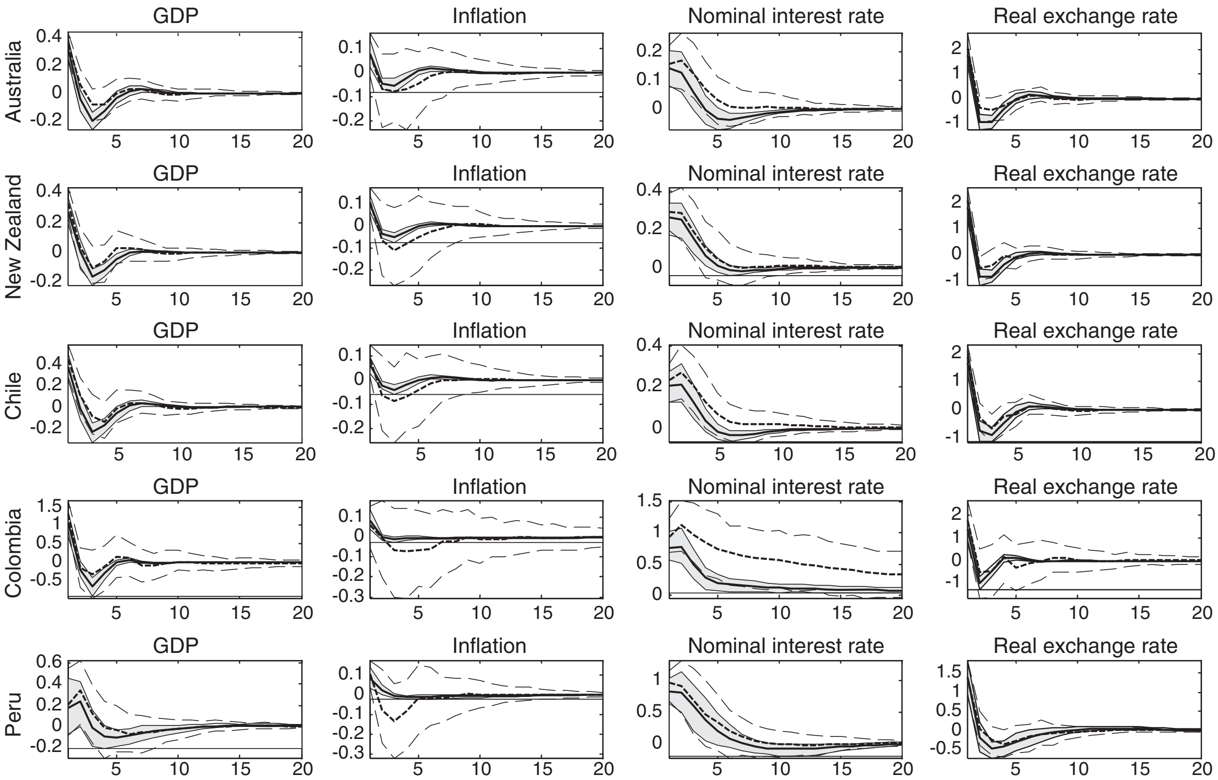

Fig. 2. Transmission of a risk-premium shock in the DSGE-VAR (dotted lines) and DSGE (plain lines) models.

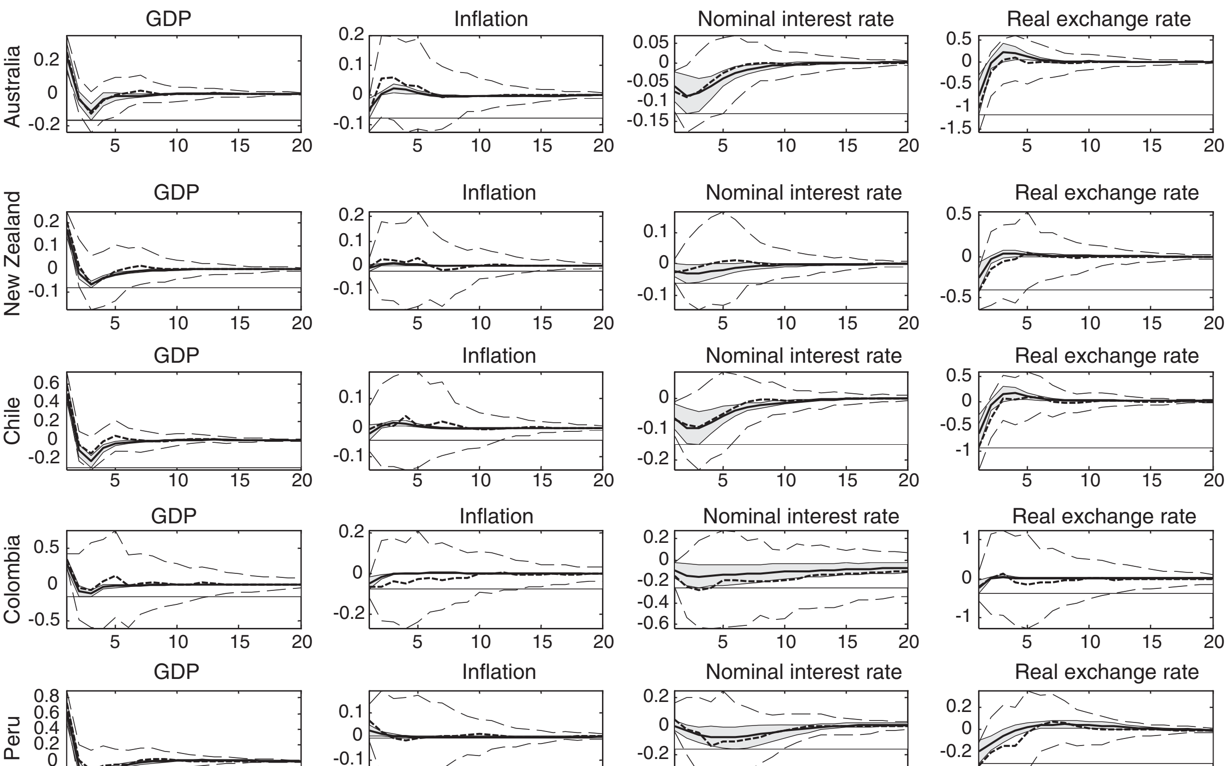

Fig. 3. Transmission of a commodity-price shock in the DSGE-VAR (dotted lines) and DSGE (solid lines) models.

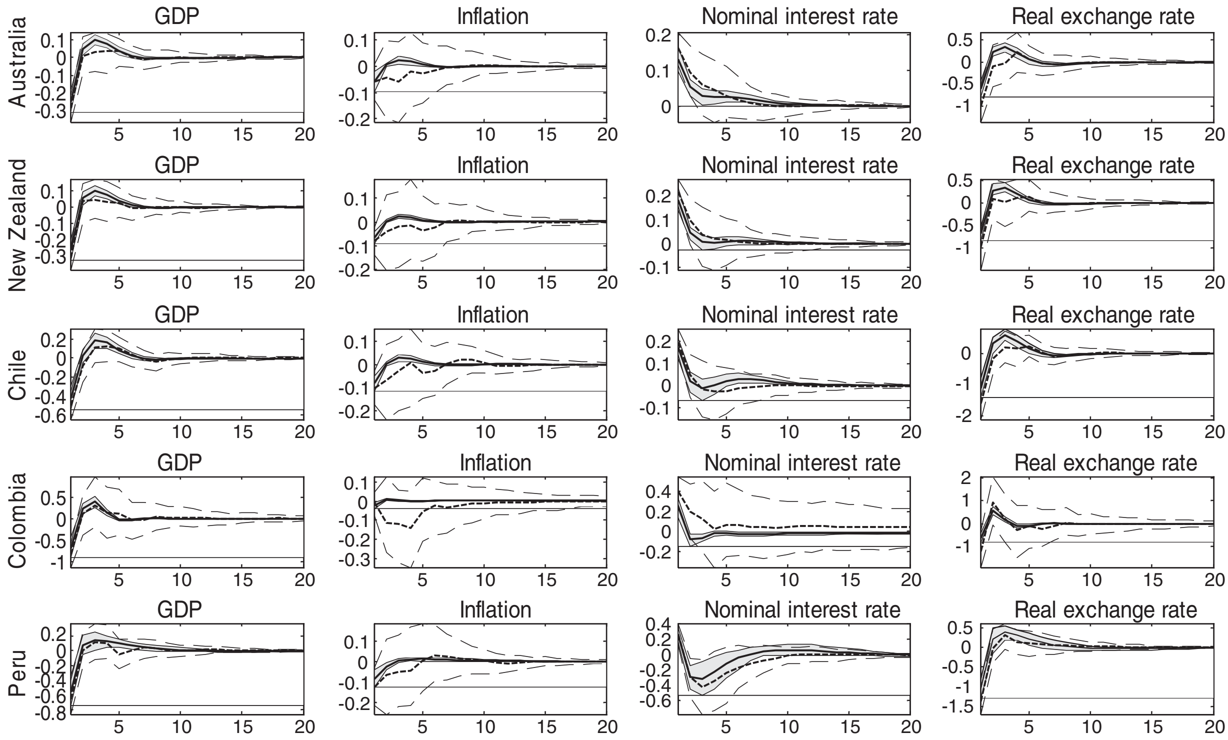

Fig. 4. Transmission of a monetary shock in the DSGE-VAR (dotted lines) and DSGE (solid lines) models.

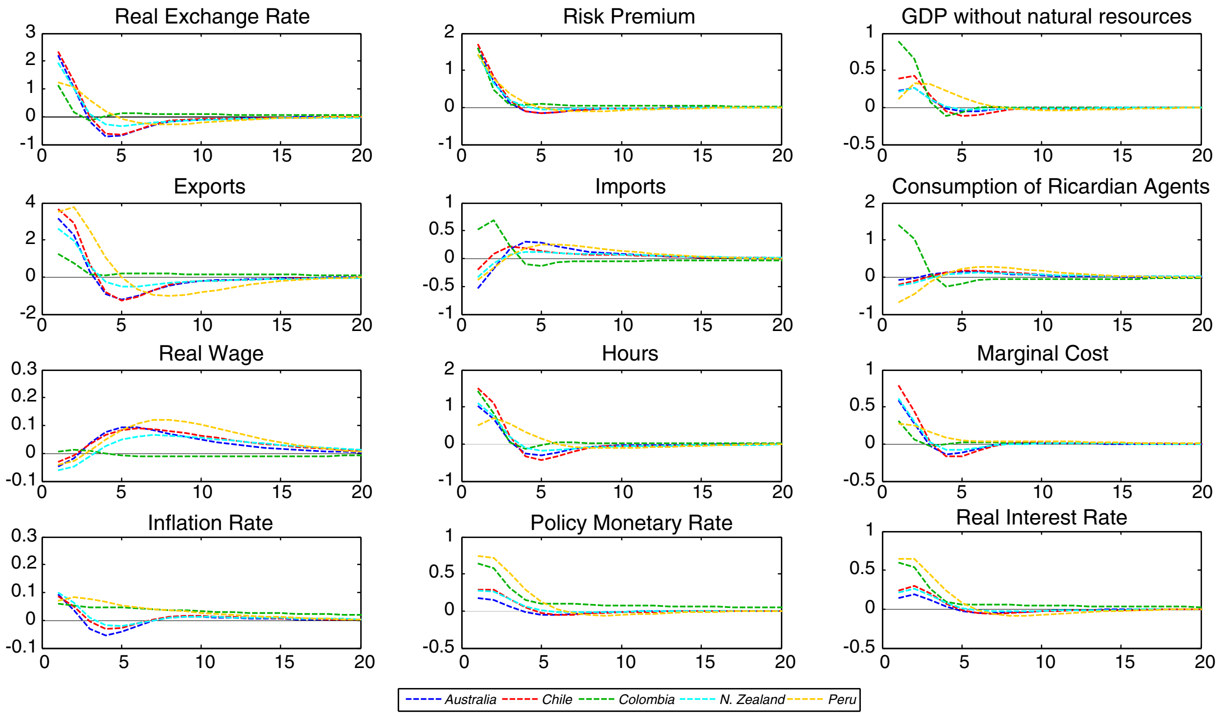

Fig. 5. Transmission of a risk-premium shock in the DSGE model: Benchmark.

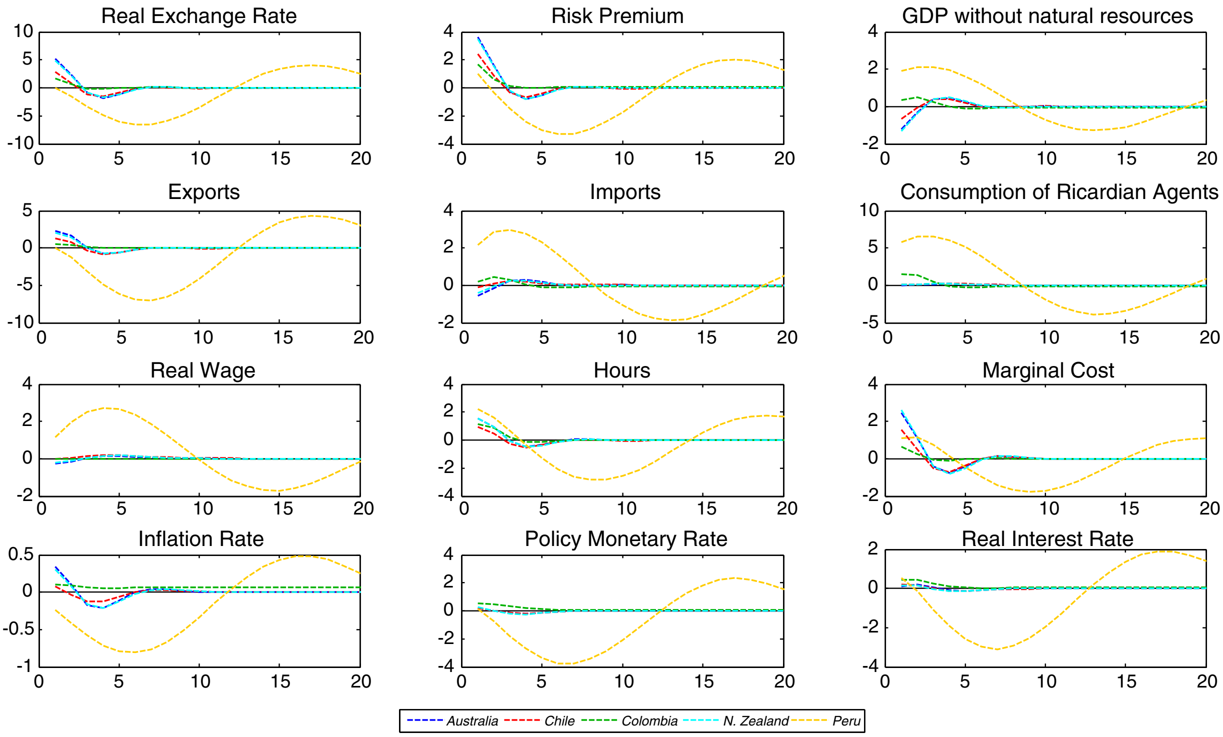

Fig. 6. Transmission of a risk-premium shock in the DSGE model: Balance sheet effect.

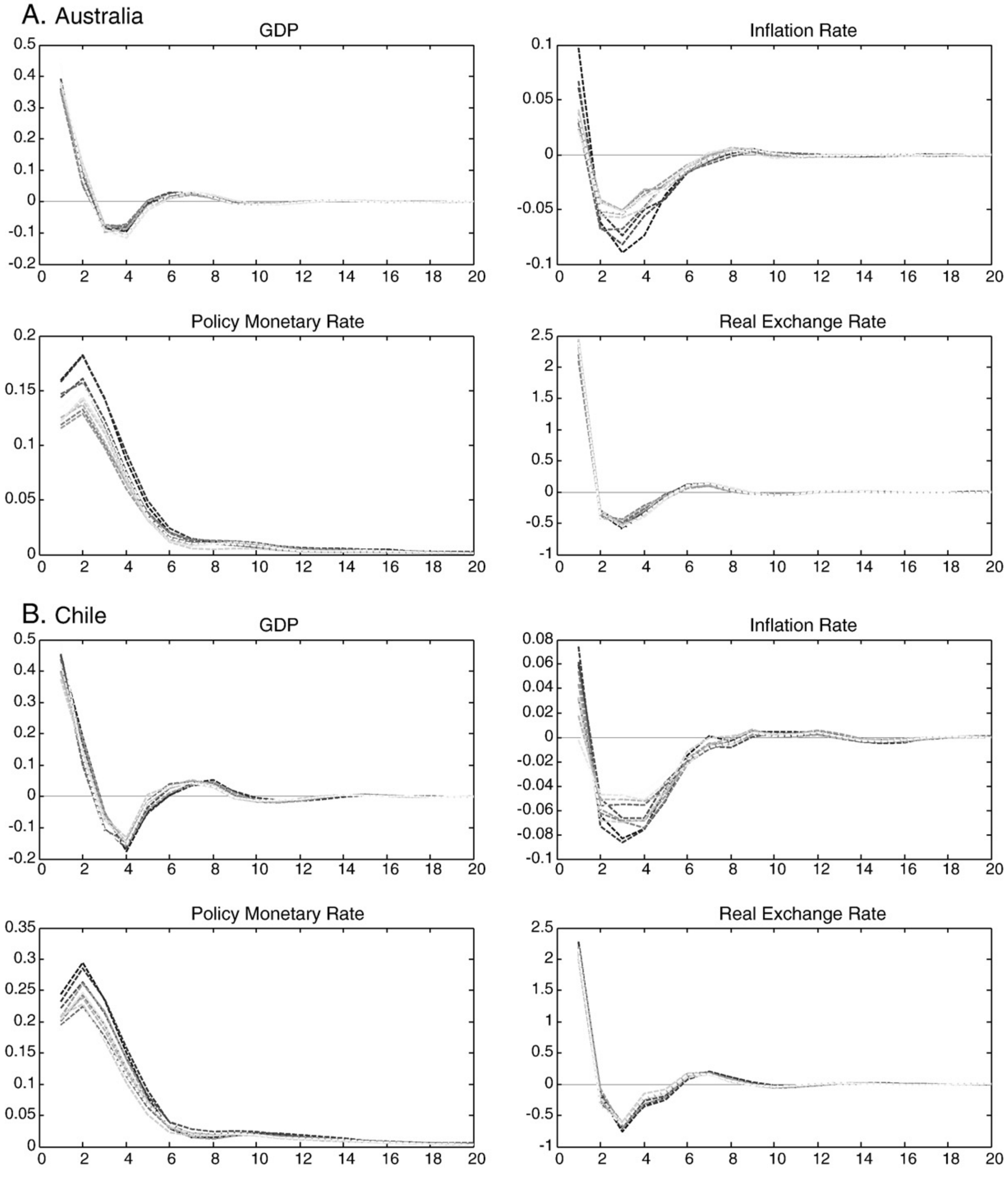

Fig. 7. Transmission of a risk-premium shock in the DSGE-VAR model. A. Australia, B. Chile, C. Colombia, D. New Zealand, and E. Peru.

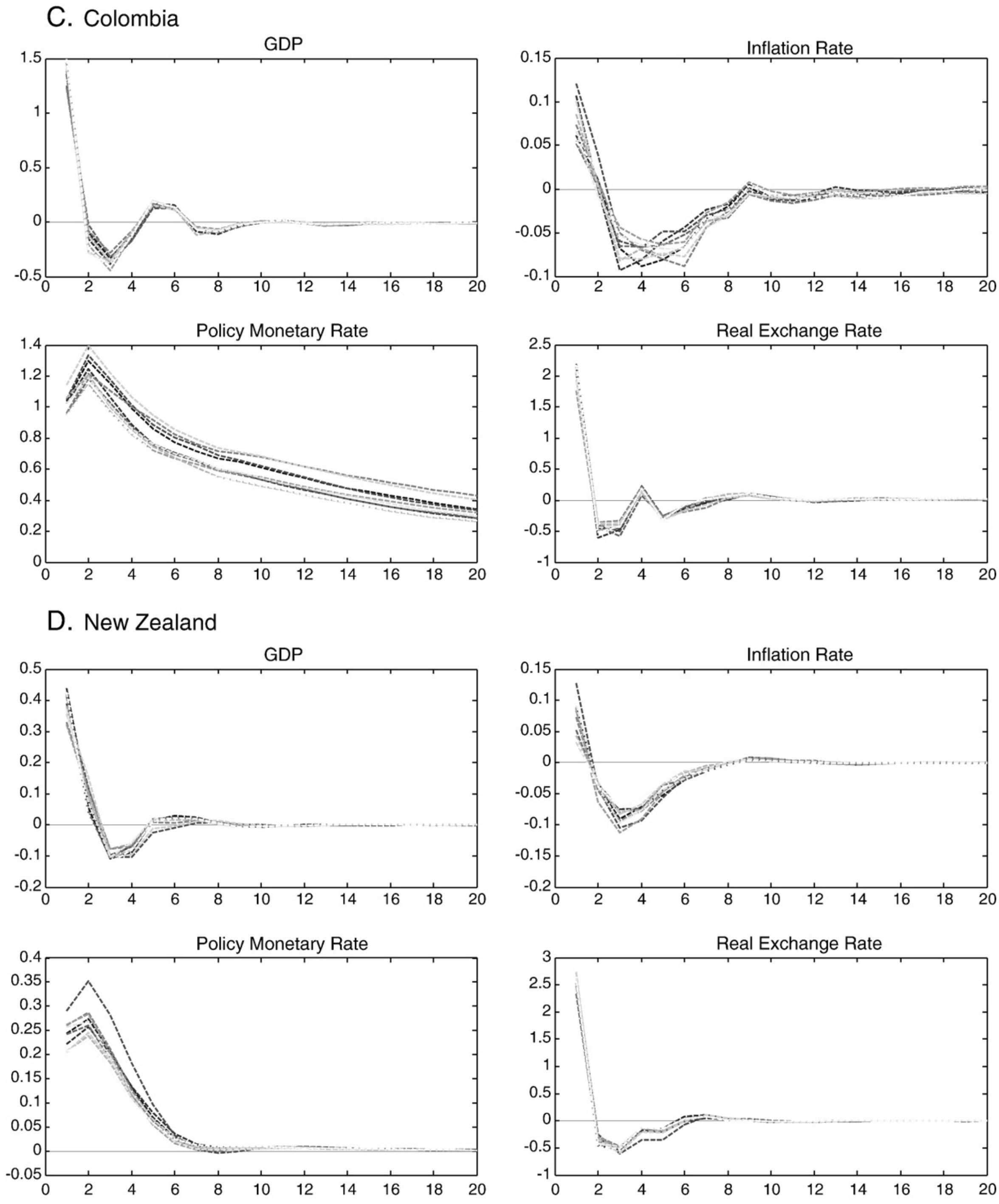

Fig. 7 (*continued*).

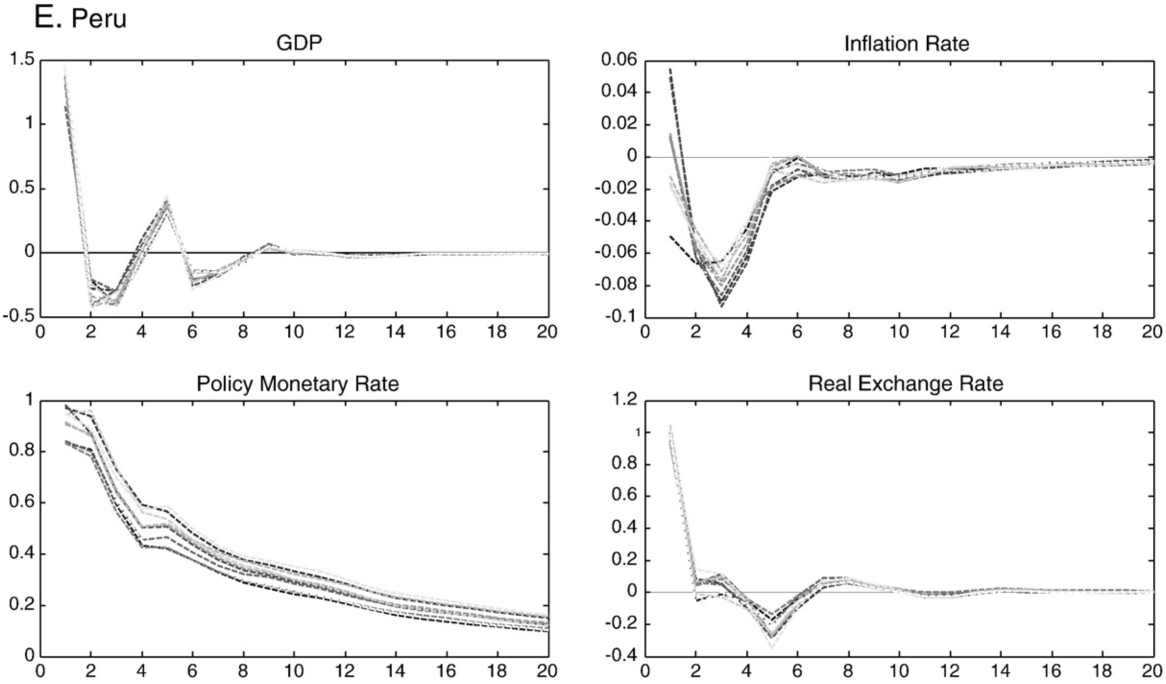

Fig. 7 (*continued*).

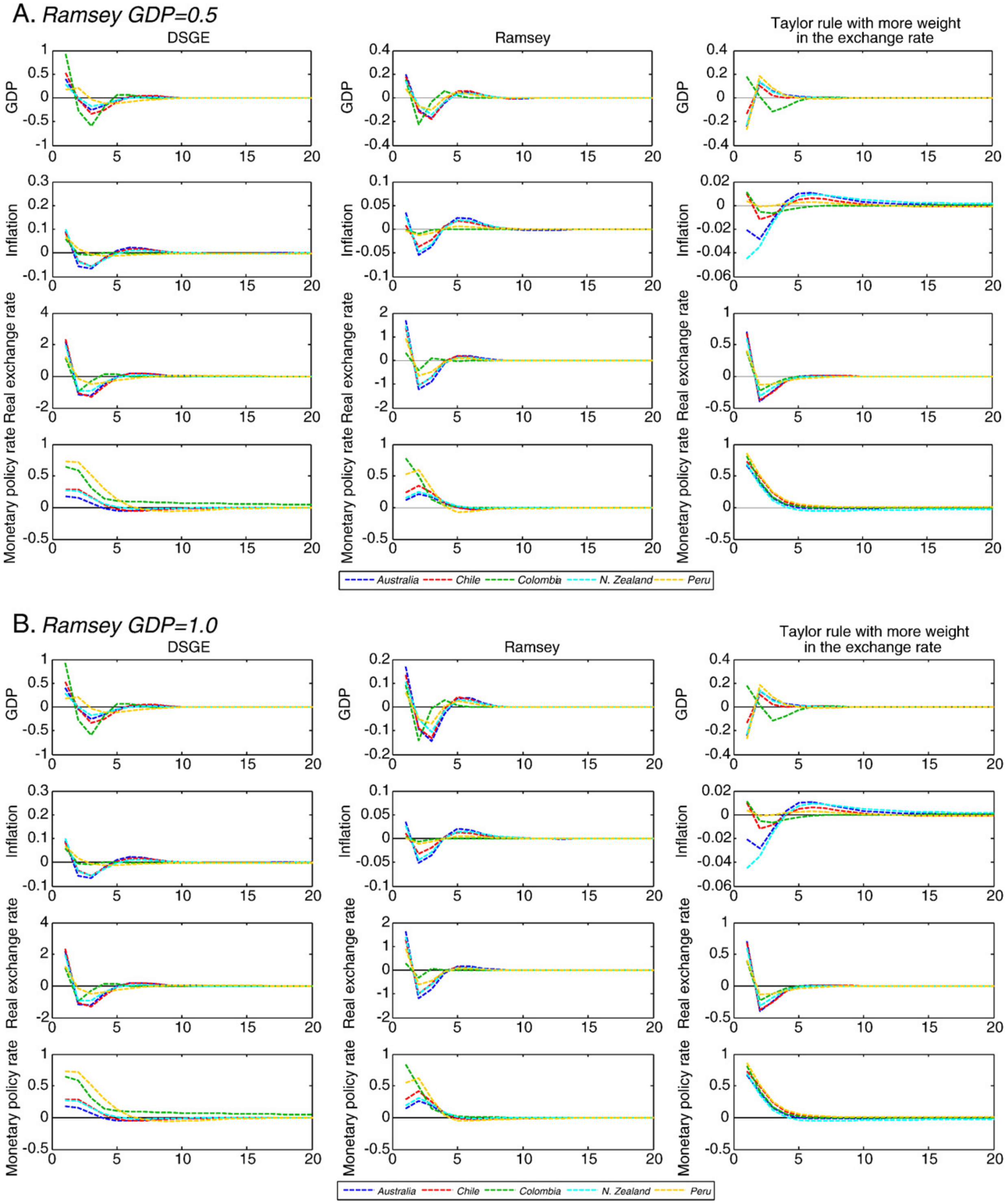

Fig. 8. Transmission of a risk-premium shock in the DSGE model: Increased weight on the exchange rate in the Taylor rule. A. Ramsey GDP = 0.5, B. Ramsey GDP = 1.0, C. Ramsey GDP = 1.5.

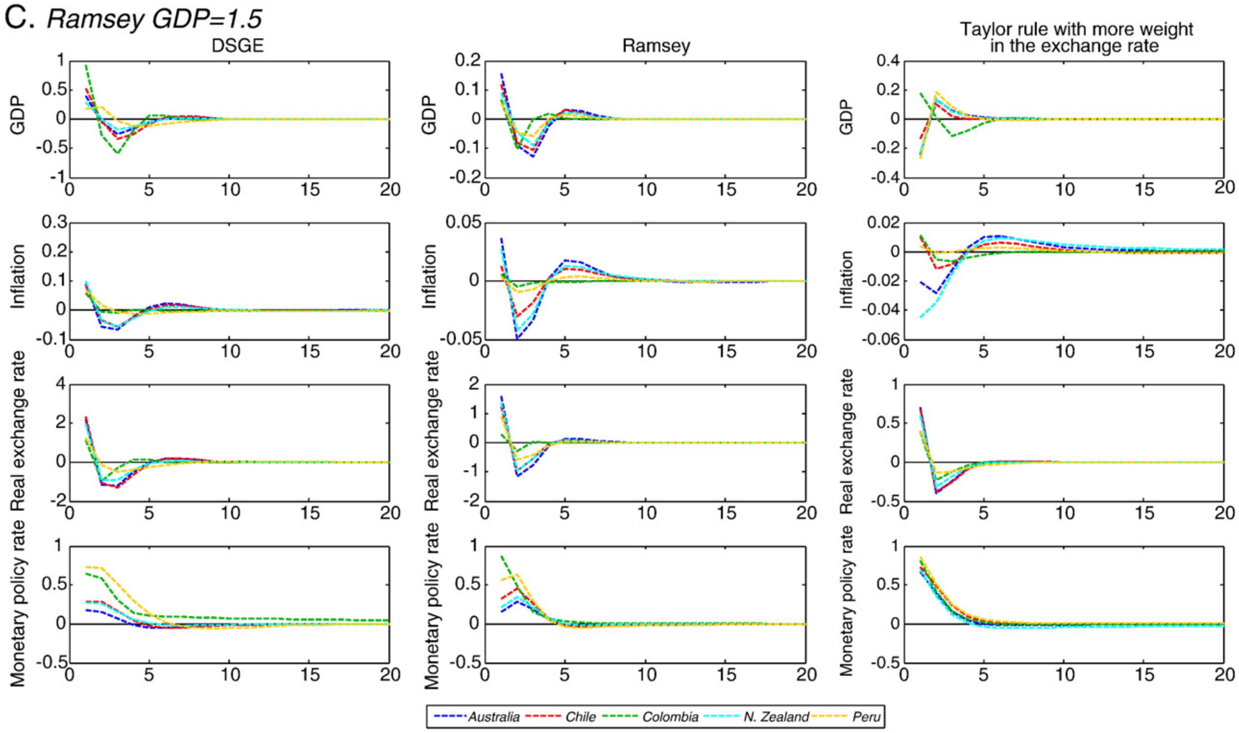

Fig. 8 (*continued*).

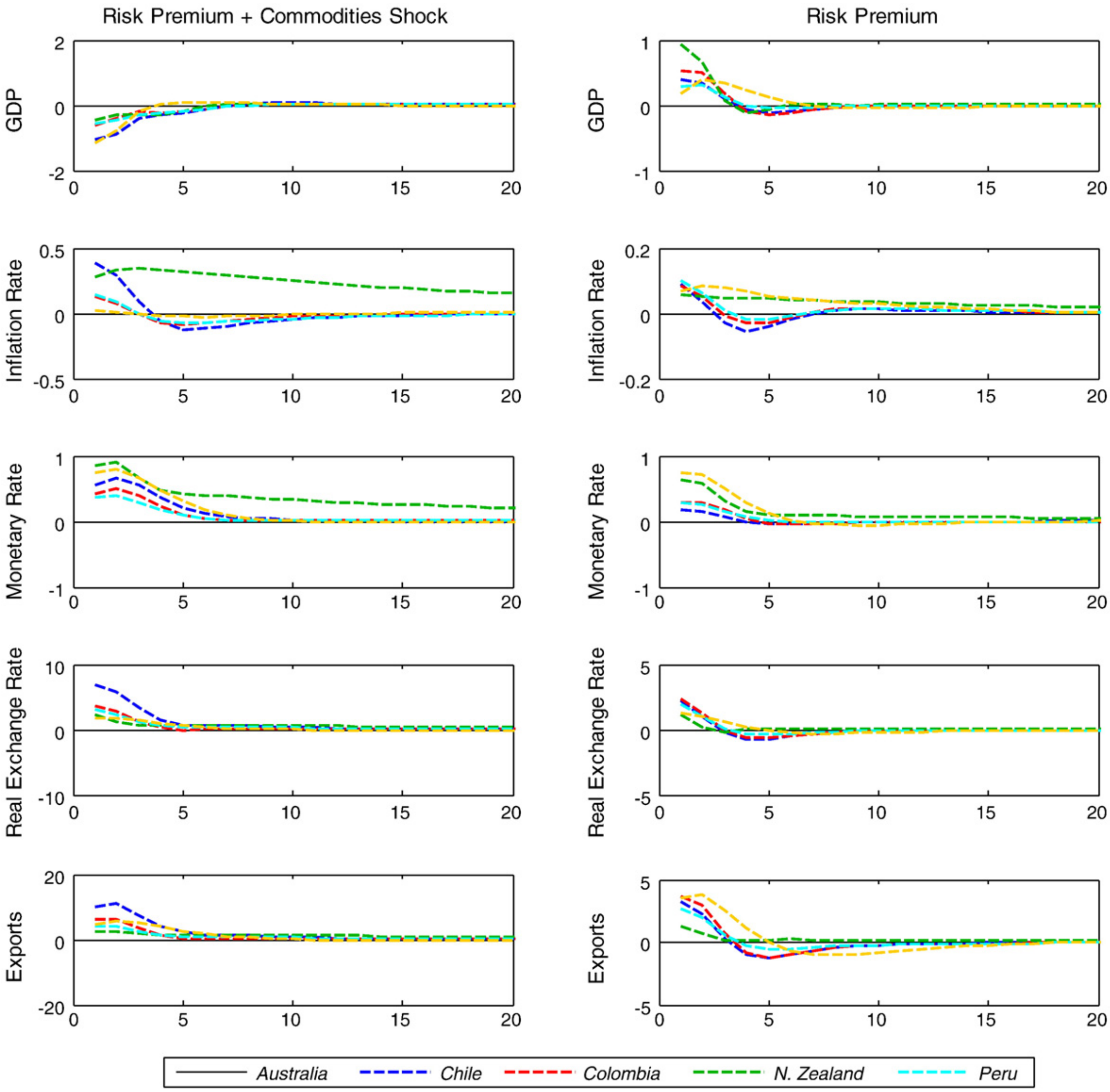

Fig. 9. Potential balance-sheet effects: A risk premium shock and a negative shock to the commodity price.

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